1 Write a Python program using recursion to generate the Fibonacci

series.

Aim

To write a Python program that uses recursion to generate the Fibonacci series up to a given number of terms.

⚙ Algorithm

1. Define a recursive function :
   - Base case: if → return 0, if → return 1.
   - Recursive case: return .
2. Ask the user for the number of terms.
3. Use a loop to call the recursive function for each term.
4. Print the series.

CODE

```
# Program to generate Fibonacci series using recursion


def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)


# Input: number of terms
terms = int(input("Enter the number of terms: "))


print(f"\nFibonacci Series up to {terms} terms:")
for i in range(terms):
```

```
    print(fibonacci(i), end=" ")
```

OUTPUT

Enter the number of terms: 10

Fibonacci Series up to 10 terms:

0 1 1 2 3 5 8 13 21 34

Conclusion

- The program uses recursion to compute Fibonacci numbers.
- Each term is calculated by calling the function recursively.
- This demonstrates how recursion can elegantly solve problems with repetitive substructure.

2 Write a program to find the largest number in a list using a function.

Aim

To write a Python program that defines a function which takes a list as input and returns the largest number in the list.

⚙ Algorithm

1. Define a function .
2. Initialize a variable with the first element of the list.
3. Traverse the list using a loop.
4. Compare each element with .
   - If the element is greater, update .
5. Return the value.
6. Call the function with a sample list and display the result.

CODE

```
# Function to find the largest number in a list
```

```python
def find_largest(lst):
    largest = lst[0]  # Step 2: assume first element is largest
    for num in lst:   # Step 3: traverse list
        if num > largest:  # Step 4: compare
            largest = num
    return largest

# Sample list
numbers = [12, 45, 78, 34, 89, 23]

# Function call
print("List:", numbers)
print("Largest number in the list:", find_largest(numbers))
```

OUTPUT

```
List: [12, 45, 78, 34, 89, 23]
Largest number in the list: 89
```

Conclusion

- The program successfully finds the largest number in a list using a function.
- It demonstrates how to traverse a list and compare elements step by step.
- This approach can be extended to find the smallest number or other statistics.